

Weekly Assignment 9: Dynamic Programming 1

1. (15 points) Run the bottom-up algorithm for rod-cutting (as described in the lecture) on the following price list with $n = 5$.

Length	1	2	3	4	5
Price	8	20	29	38	48

Clearly include the following items in your solutions:

- The constructed function f as a table.
 - The computations used to fill right-most column in table $f(5)$.
 - The maximum possible total value of this rod cutting instance.
2. (25 points) Suppose you are going on a biking trip during your holiday. You start on the road at kilometer post 0. Along the way there are n hotels, at kilometer posts $h_1 < h_2 < \dots < h_n$, where each h_i is measured from the starting point. The only places you are allowed to stop are at these hotels, but you can choose which of the hotels you stop at. You must stop at the final hotel (at distance h_n), which is your destination. Ideally you would like to travel 200 kilometer a day, but this may not be possible (depending on the distance between the hotels). If you travel m kilometers during a day, the penalty for that day is $(200 - m)^2$. You want to plan your trip so as to minimize the total penalty, that is, the sum, over all travel days, of the daily penalties.

Use dynamic programming to give an efficient algorithm that determines the optimal sequence of hotels at which to stop. Specify the recursion equations on which your solution is based, explain why these equations are correct, describe a bottom - up implementation based on the recursion equations, and analyze the time complexity of your algorithm. **Hint:** Let p be an array where $p[j]$ is the minimum penalty required to travel to hotel j .

3. (30 points) Let $G = (V, E)$ be a directed graph with nodes $v_1, \dots, v_n$. We say that G is a *line-graph* if it has the following properties:
 1. Each edge goes from a node with a lower index to a node with a higher index. That is, every directed edge has the form (v_i, v_j) with $i < j$.
 2. Each node except v_n has at least one edge leaving it. That is, for every node v_i , $i = 1, 2, \dots, n - 1$, there is at least one edge of the form (v_i, v_j) .

The length of a path is the number of edges in it. The goal in this question is to solve the following algorithmic problem (see Figure 1 for an example):

For a line-graph G , find the length of the longest path that begins at v_1 and ends at v_n .

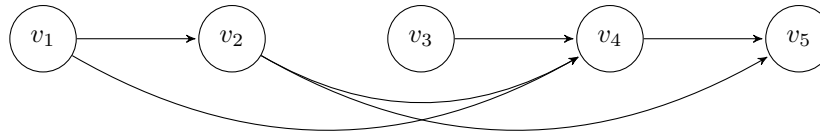


Figure 1: The correct answer for this line-graph is 3: the longest path from v_1 to v_n uses the three edges (v_1, v_2) , (v_2, v_4) , and (v_4, v_5) .

- (a) Show that the following algorithm does not correctly solve this problem, by giving an example of a line-graph on which it does not return the correct answer. In your example, say what the correct answer is and also what the algorithm finds.

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Set  $w = v_1$ 
Set  $L = 0$ 
while there is an edge out of the node  $w$  do
    Choose the edge  $(w, v_j)$  for which  $j$  is as small as possible
    Set  $w = v_j$ 
    Increase  $L$  by 1
end while
return  $L$  as the length of the longest path

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- (b) Give an algorithm that takes a line graph G and returns the length of the longest path that begins at v_1 and ends at v_n . The running time of your algorithm should be polynomial in n . Argue that your algorithm works correctly, and include a brief analysis of the running time.
4. (30 points) Use dynamic programming to give an $\mathcal{O}(nt)$ algorithm for the following task.
- Input:* A list of n positive integers $a_1, a_2, \dots, a_n$; a positive integer t .
- Output:* Does some subset of the a_i 's add up to t ? (You can use each a_i at most once.)
- Give a set of recursion equations that define the (Boolean) value of the solution, explain why these equations hold, and base your algorithm on them.